

Forecasting Monthly Cardiovascular Prescriptions in Australia: A SARIMA Time Series Approach Using PBS Medicare Data

MATH1318 Time Series Analysis – Final Project Report

Prepared by:

Logadharshini Karuppasamy (S4148604)

RMIT University,

School of Science

Table of Contents

Table of Contents	2
1. Introduction	4
1.1 Background and Motivation	4
1.2 Dataset Description	4
1.3 Research Objective	5
1.4 Report Structure	5
2. Descriptive and Exploratory Analysis	5
2.1 Time Series Plot	5
2.2 Descriptive Statistics	7
2.3 Decomposition	7
2.4 Seasonal Plot.....	9
2.5 ACF and PACF of the Original Series.....	10
3. Stationarity Analysis and Data Transformation	11
3.1 Stationarity Tests on the Original Series.....	11
3.2 Determining the Order of Differencing	11
3.3 Log Transformation.....	12
3.4 Differencing.....	13
3.5 ACF and PACF After Differencing	14
3.6 Stationarity Tests After Differencing	14
4. Model Specification	15
4.1 Extended Autocorrelation Function (EACF).....	15
4.2 BIC Table (armasubsets).....	16
4.3 Candidate Models.....	17
5. Model Fitting, Selection and Parameter Estimation.....	18
5.1 Fitting the Candidate Models	18
5.2 Model Comparison.....	18
5.3 Verification with auto.arima()	19
5.4 Parameter Estimates for the Selected Model	20
6. Diagnostic Checking.....	20
6.1 Residual Plots.....	21
6.2 Ljung-Box Test for Residual Autocorrelation	22

6.3 Shapiro-Wilk Normality Test	23
6.4 Overall Diagnostic Conclusion	23
7. Forecasting	24
7.1 10-Month Forecast Table.....	24
7.2 Forecast Plot.....	25
7.3 Discussion of Forecast Results.....	25
8. Conclusion	26
8.1 Summary of Findings	26
8.2 Limitations and Future Work	27
8.3 Practical Implications	27
9. References.....	28
10. Appendix A: Complete R Code	29

1. Introduction

1.1 Background and Motivation

Cardiovascular disease (CVD) remains one of the leading causes of morbidity and mortality in Australia, and the long-term management of these conditions relies heavily on the consistent supply and subsidisation of cardiovascular medicines. The Australian Government's Pharmaceutical Benefits Scheme (PBS) subsidises the cost of a wide range of prescription medicines, including cardiovascular drugs, for concession card holders. Understanding how the volume of subsidised cardiovascular prescriptions changes over time, and being able to forecast future demand, is valuable for health policy planning, pharmaceutical supply chain management, and budget forecasting at a national level.

This report applies time series analysis techniques covered in MATH1318 Time Series Analysis to model and forecast the monthly number of cardiovascular prescription scripts dispensed under the PBS Co-payments (Concessional) scheme. A Seasonal Autoregressive Integrated Moving Average (SARIMA) modelling framework is used, with model orders identified using the Extended Autocorrelation Function (EACF) and the Bayesian Information Criterion (BIC) table, consistent with the methodology taught in class.

1.2 Dataset Description

The dataset used in this study is sourced from the `tsibbledata` R package, which provides historical PBS dispensing data for Australia. The data was filtered to extract monthly cardiovascular (ATC1 = "C") prescription scripts dispensed under the Co-payments scheme for Concessional patients. The filtered records were aggregated by month to produce a single univariate monthly time series.

The resulting series, referred to throughout this report as `cardio_ts`, consists of 204 monthly observations spanning from July 1991 to June 2008 (17 years). Each observation represents the total number of cardiovascular prescription scripts dispensed under the specified PBS category for that month.

1.3 Research Objective

The primary objective of this report is to identify an appropriate SARIMA model for the cardio_ts series and use it to forecast the monthly number of cardiovascular prescriptions for the 10 months following the end of the observed data, namely July 2008 to April 2009. The specific research question addressed is:

“What will the monthly number of cardiovascular prescription scripts dispensed under the PBS Co-payments (Concessional) scheme be for the next 10 months (July 2008 – April 2009), based on historical patterns from July 1991 to June 2008?”

1.4 Report Structure

The remainder of this report is structured as follows. Section 2 presents a descriptive and exploratory analysis of the time series, including decomposition and seasonal characteristics. Section 3 examines the stationarity of the series and describes the transformation and differencing applied to achieve stationarity. Section 4 details the model specification process using the EACF table and the BIC table from the armasubsets function. Section 5 presents the fitting, comparison, and selection of candidate SARIMA models, along with parameter estimates for the selected model. Section 6 reports the diagnostic checking of model residuals. Section 7 presents the 10-month forecast. Section 8 concludes the report with a summary of findings, limitations, and recommendations. An acknowledgement of generative AI tool use is provided in Section 9, followed by references and an appendix containing the complete R code used for the analysis.

2. Descriptive and Exploratory Analysis

2.1 Time Series Plot

Figure 1 displays the time plot of the monthly cardiovascular prescription series from July 1991 to June 2008. The series was first converted into an R ts object with a frequency of 12 (monthly data) before plotting.

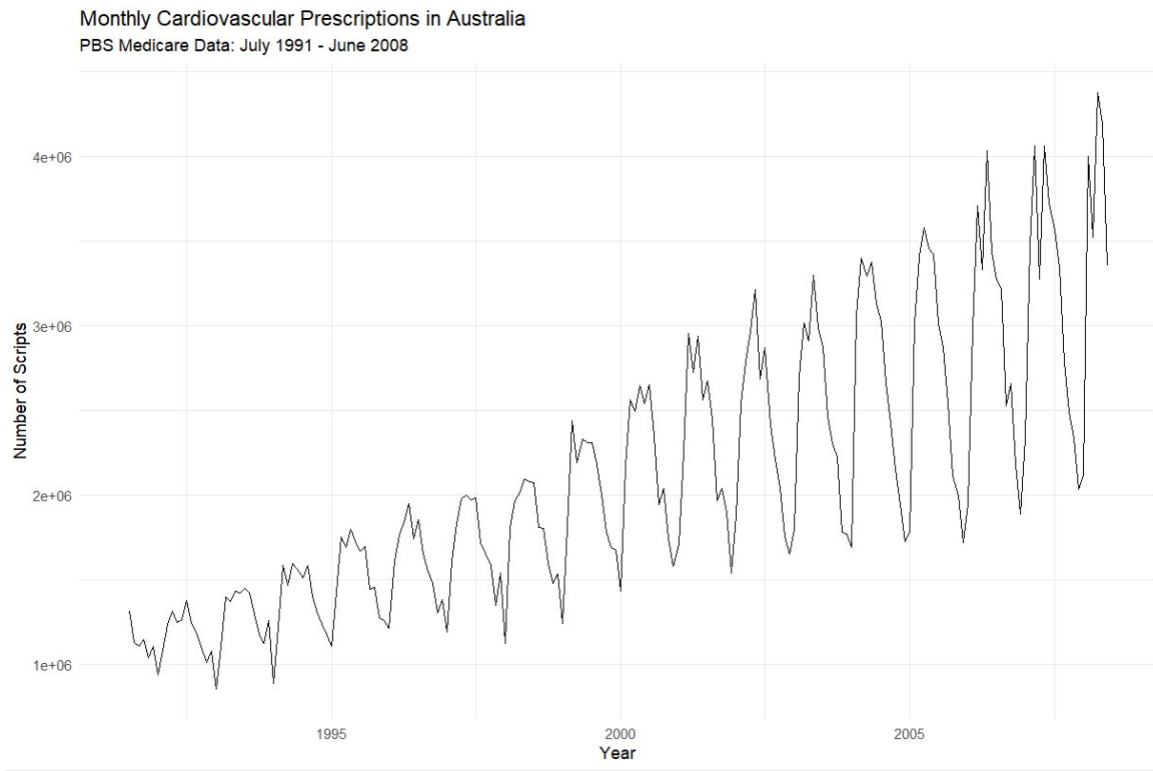


Figure 1. Time plot of monthly cardiovascular prescription scripts, July 1991 – June 2008.

The time plot reveals three key characteristics of the series:

- **Trend:** There is a strong, persistent upward trend across the full 17-year period, with the number of monthly scripts increasing from approximately 855,000 in 1991 to over 4.3 million by 2008. This is consistent with Australia's ageing population and the increasing prevalence of cardiovascular disease diagnoses.
- **Seasonality:** A clear and consistent annual cycle is visible, with prescription volumes peaking around February to April each year and reaching a trough around July to September.
- **Changing variance:** The amplitude of the seasonal fluctuations increases as the level of the series increases over time. This pattern – where the seasonal variation is proportional to the level of the series – is characteristic of a multiplicative time series and motivates the use of a log transformation (Section 3.3) and a multiplicative decomposition (Section 2.2).

2.2 Descriptive Statistics

Summary statistics for the `cardio_ts` series are presented in Table 1.

Statistic	Value (scripts)
Minimum	855,398
1st Quartile	1,475,258
Median	1,922,140
Mean	2,101,152
3rd Quartile	2,647,972
Maximum	4,377,361
Standard Deviation	791,127.9
Number of observations (n)	204

Table 1. Descriptive statistics for the monthly cardiovascular prescription series (`cardio_ts`).

The large gap between the minimum and maximum values, combined with a standard deviation of approximately 791,128, reflects both the strong upward trend and the substantial seasonal swings present in the data.

2.3 Decomposition

To better understand the underlying structure of the series, a classical multiplicative decomposition was performed using the `decompose()` function in R, separating the series into trend, seasonal, and random (remainder) components. A multiplicative decomposition was chosen over an additive one because, as noted in Section 2.1, the magnitude of the seasonal fluctuations grows in proportion to the level of the series.

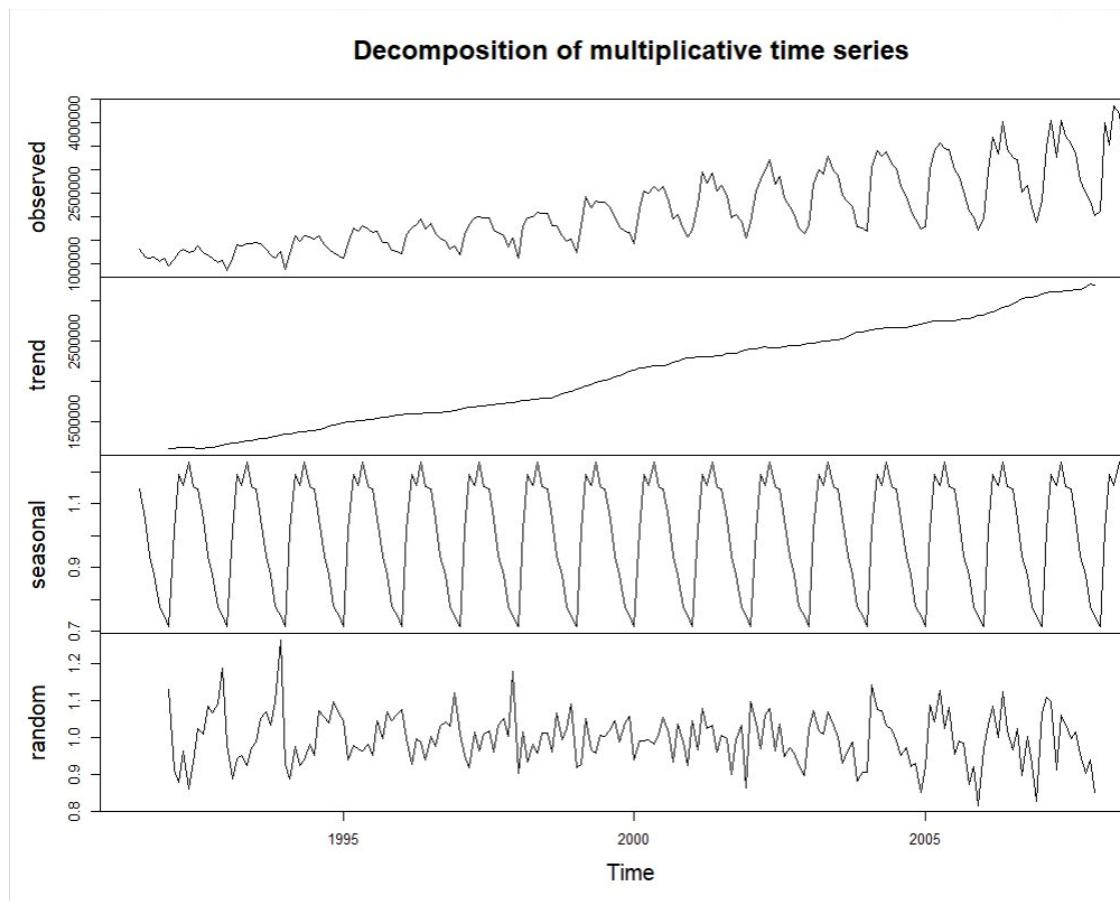


Figure 2. Multiplicative decomposition of the cardiovascular prescription series into trend, seasonal, and random components.

The decomposition in Figure 2 confirms the visual impressions from the time plot:

- **Trend component:** A smooth, steadily increasing trend, rising from approximately 855,000 to around 4.4 million scripts over the 17-year period. This long-term growth is consistent with Australia's ageing population and rising rates of cardiovascular diagnosis and treatment.
- **Seasonal component:** A strongly repeating annual pattern with seasonal indices ranging from approximately 0.7 to 1.2. Values above 1 correspond to months where prescription volumes are typically higher than the trend would suggest (autumn/winter months), while values below 1 correspond to months with lower-than-trend volumes (the mid-year period).
- **Random component:** The remainder component fluctuates around 1.0 with relatively stable variance, indicating that the multiplicative decomposition has adequately captured the systematic trend and seasonal structure of the series.

2.4 Seasonal Plot

Figure 3 presents a seasonal plot of the series, in which each year is plotted as a separate line across the months of the year (January to December). This view makes the seasonal pattern, and how it has evolved over time, particularly easy to interpret.

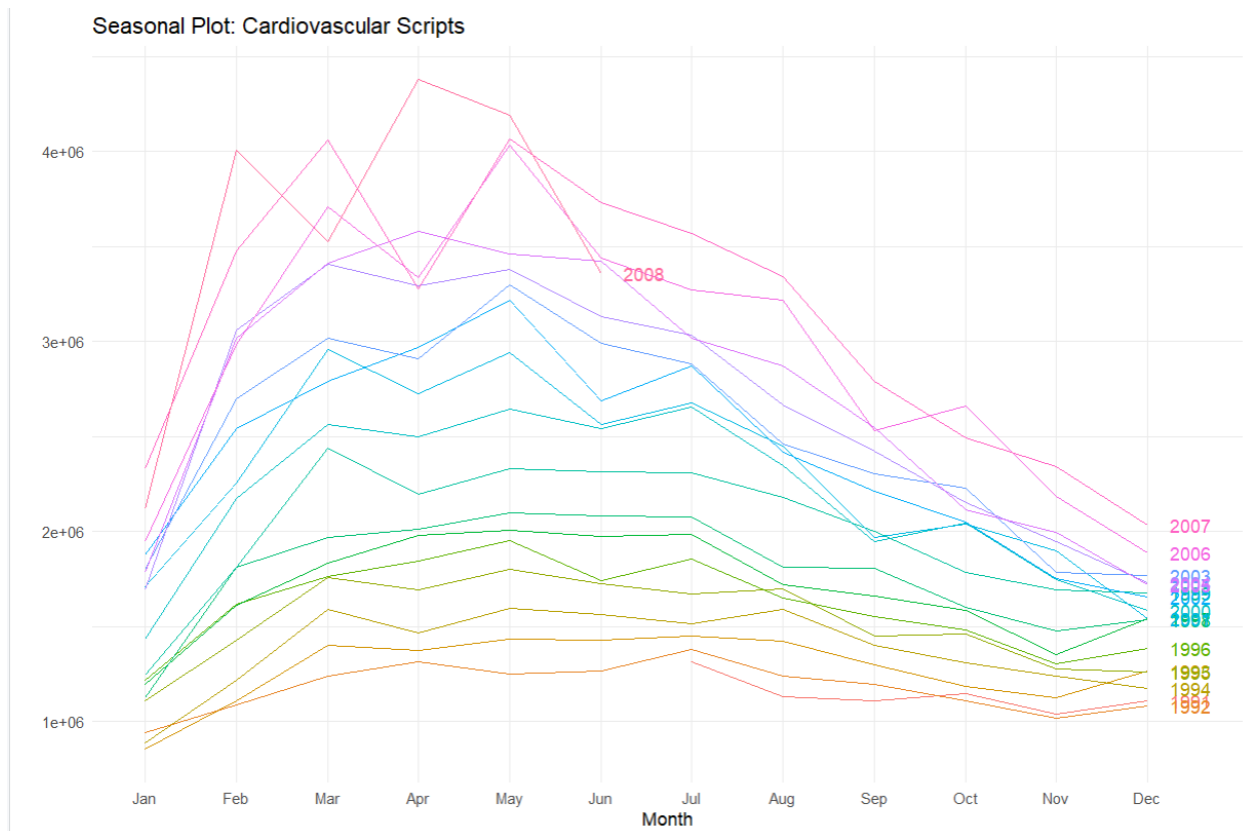


Figure 3. Seasonal plot of monthly cardiovascular prescription scripts by year.

The seasonal plot confirms that, in almost every year of the observed period, cardiovascular prescription volumes rise from a low point in January, peak somewhere between February and May (a pattern consistent with increased cardiovascular events and prescription renewals during the cooler autumn and winter months in Australia), and then decline towards a trough in the months of July to September, before beginning to rise again towards the end of the year. The overall upward shift of the lines from the earliest years (bottom of the plot, in orange/yellow tones) to the most recent years (top of the plot, in pink/magenta tones, including 2008) again illustrates the long-run upward trend identified in Sections 2.1 and 2.2.

2.5 ACF and PACF of the Original Series

Figure 4 shows the sample autocorrelation function (ACF) and partial autocorrelation function (PACF) of the original (untransformed, undifferenced) series `cardio_ts`, computed up to 48 lags (four years).

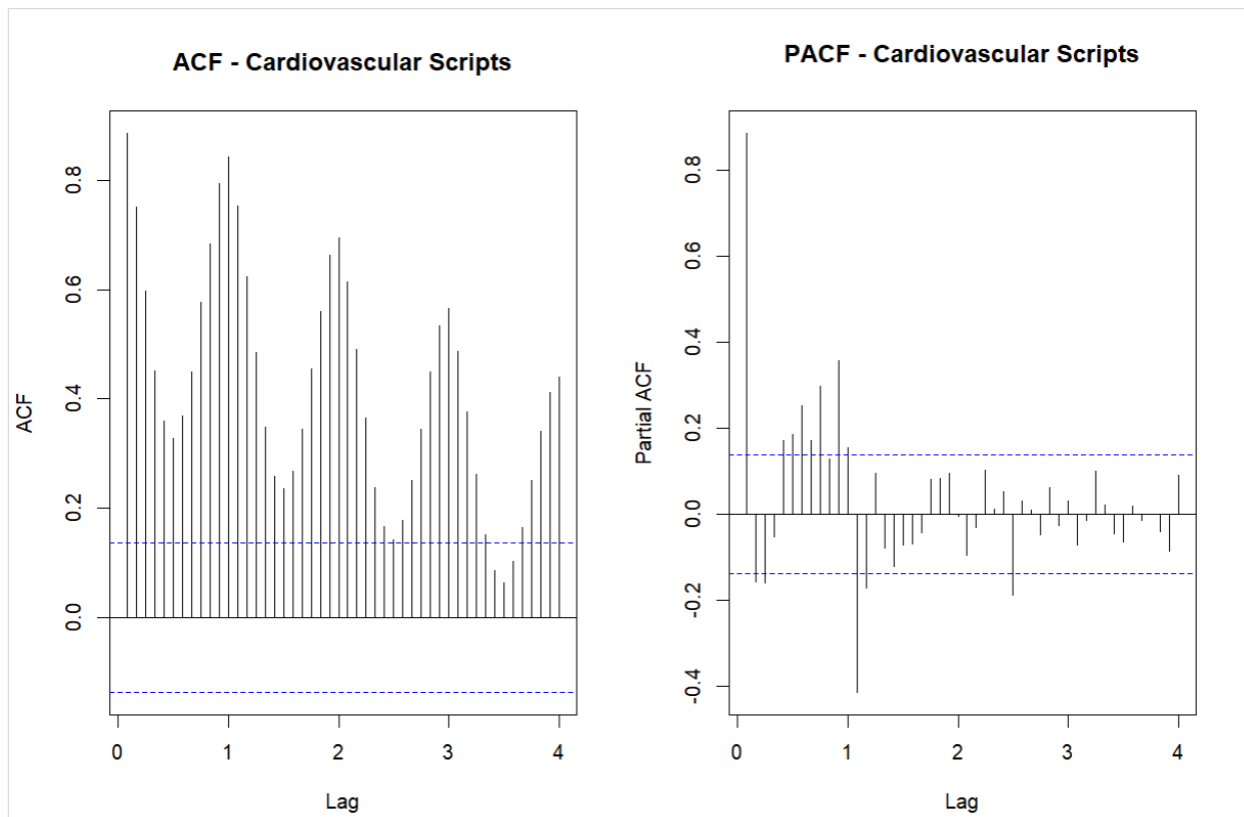


Figure 4. ACF (left) and PACF (right) of the original cardiovascular prescription series.

The ACF in Figure 4 decays very slowly and remains well above the significance bounds for many lags, which is a classic indicator of a non-stationary series with a strong trend. In addition, a repeating ripple pattern with a period of approximately 12 lags is visible in the ACF, reflecting the seasonal structure identified in Sections 2.1–2.4. The PACF shows a very large spike at lag 1, with a number of further significant spikes at low lags, including a large negative spike around lag 12, again consistent with both trend and seasonal dependence. Together, these plots indicate that the series is non-stationary in both its level and its seasonal structure, and that transformation and differencing will be required before an ARIMA-type model can be fitted. This is examined formally in Section 3.

3. Stationarity Analysis and Data Transformation

3.1 Stationarity Tests on the Original Series

Before fitting an ARIMA-type model, the series must be stationary, meaning its mean, variance, and autocorrelation structure do not change over time. Two complementary formal tests were applied to the original series `cardio_ts`: the Augmented Dickey-Fuller (ADF) test, where the null hypothesis is that the series is non-stationary, and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, where the null hypothesis is that the series is stationary. The results are summarised in Table 2.

Test	Statistic	p-value	Conclusion
ADF (original series)	Dickey-Fuller = -13.07 (lag order = 5)	0.01	Reject H0 of non-stationarity
KPSS (original series)	KPSS Level = 3.3619	0.01	Reject H0 of stationarity

Table 2. ADF and KPSS test results for the original series `cardio_ts`.

At first glance, these two results appear contradictory: the ADF test rejects non-stationarity ($p = 0.01$), while the KPSS test rejects stationarity ($p = 0.01$). This apparent contradiction is common for series that exhibit strong seasonality. The ADF test, in its standard form, primarily tests for a unit root associated with the overall trend and does not explicitly account for seasonal unit roots, whereas the KPSS test is more sensitive to the combination of trend and seasonal non-stationarity present in the series. Taken together with the clear seasonal pattern and increasing variance observed in Section 2, these results indicate that the series requires both a variance-stabilising transformation and differencing (both seasonal and/or regular) before it can be considered stationary.

3.2 Determining the Order of Differencing

The `ndiffs()` and `nsdiffs()` functions from the `forecast` package were used to formally determine the number of regular and seasonal differences required to achieve stationarity. The results were:

- **`ndiffs(cardio_ts) = 1`** – one regular (non-seasonal) difference is recommended.
- **`nsdiffs(cardio_ts) = 1`** – one seasonal difference (at lag 12) is recommended.

These results, together with the visual evidence of changing variance from Section 2, guided the preprocessing steps described in the following subsections: a log transformation to stabilise the variance, followed by a seasonal difference and a regular difference applied during preprocessing to confirm stationarity.

3.3 Log Transformation

Because the seasonal fluctuations in the series increase in proportion to its level (a multiplicative pattern, as discussed in Section 2.1), a natural logarithm transformation was applied to the series to stabilise the variance:

$$\text{cardio_ts_log} = \log(\text{cardio_ts})$$

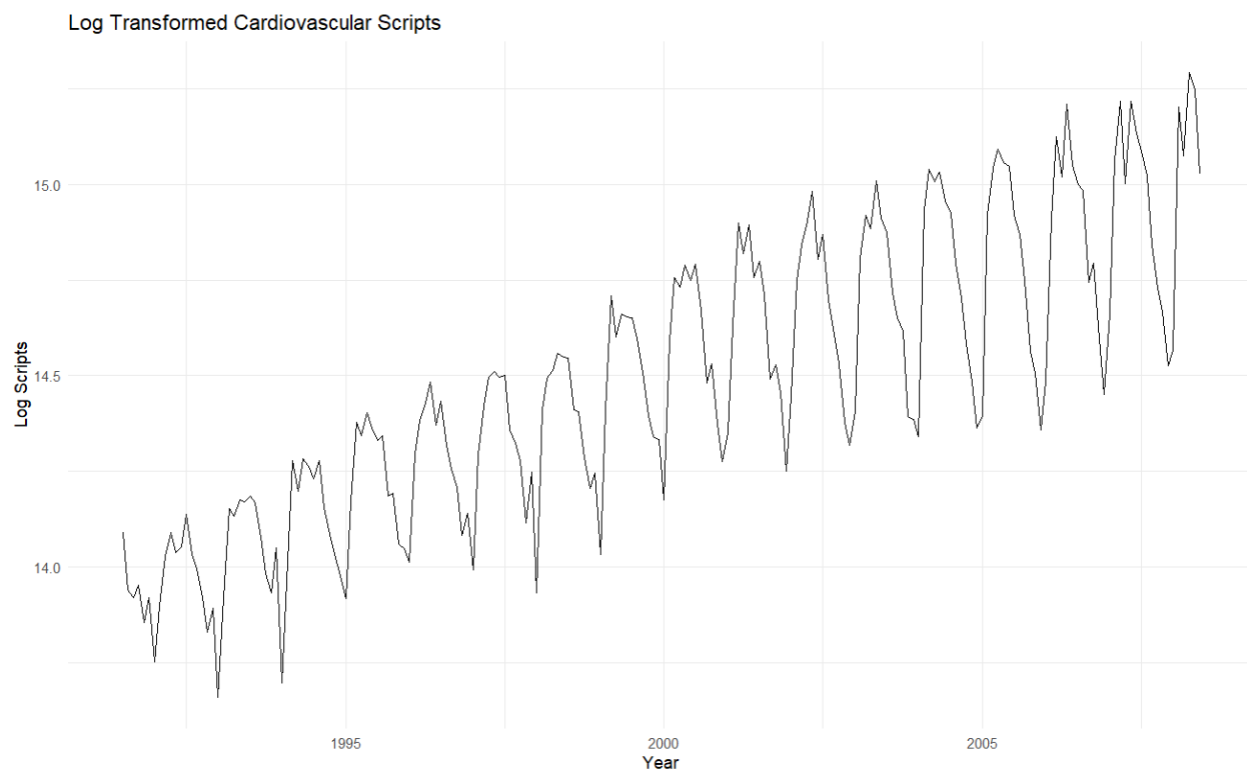


Figure 5. Time plot of the log-transformed cardiovascular prescription series (*cardio_ts_log*).

As shown in Figure 5, the log transformation has been effective in stabilising the variance: the amplitude of the seasonal fluctuations is now much more consistent across the full length of the series, compared with the original series in Figure 1. The strong upward trend and annual seasonal pattern remain clearly visible, indicating that differencing is still required to achieve stationarity.

3.4 Differencing

Following the recommendations from `ndiffs()` and `nsdiffs()` in Section 3.2, the log-transformed series was differenced during preprocessing as follows. First, a seasonal difference at lag 12 was applied to remove the annual seasonal pattern:

```
cardio_diff = diff(cardio_ts_log, lag = 12)
```

Second, a regular (first-order) difference was applied to the seasonally differenced series to remove any remaining trend:

```
cardio_diff2 = diff(cardio_diff, differences = 1)
```

The resulting twice-differenced series, `cardio_diff2`, is plotted in Figure 6.

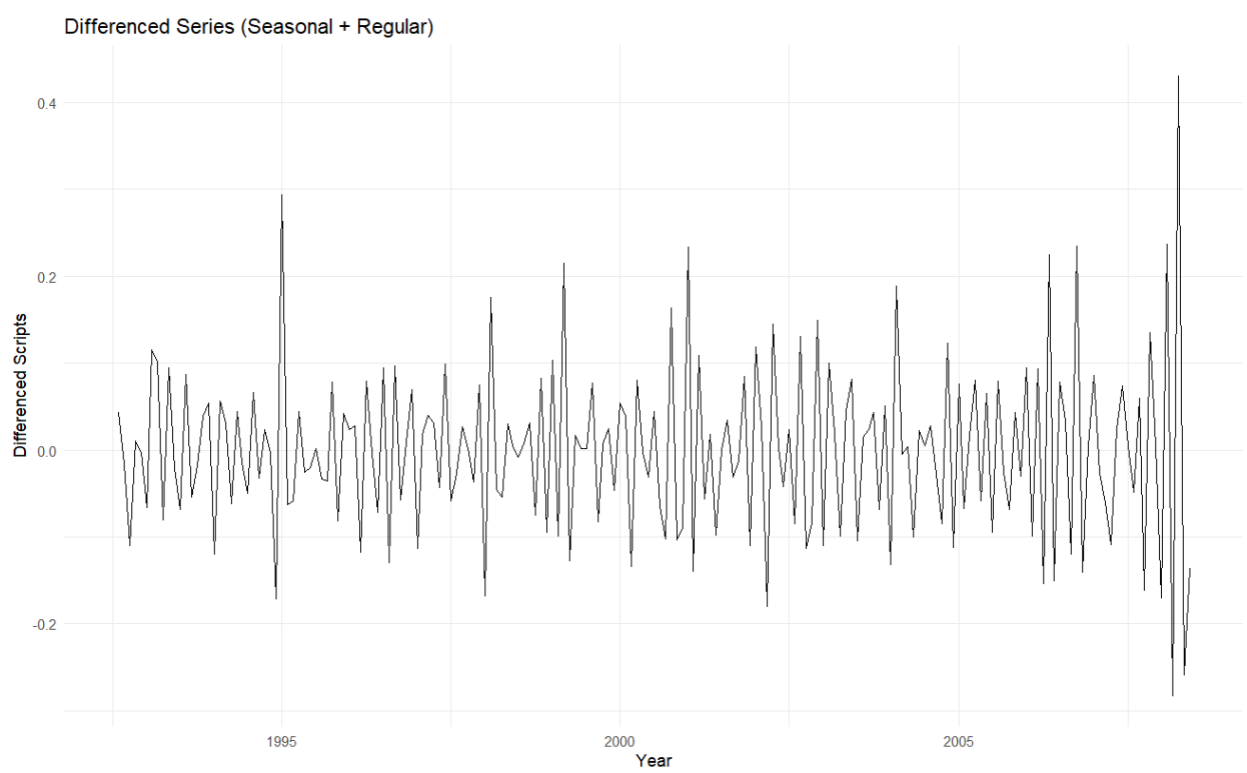


Figure 6. Time plot of the seasonally and regularly differenced log series (`cardio_diff2`).

Figure 6 shows a series that fluctuates around zero with no obvious trend or remaining seasonal pattern, suggesting that the combination of log transformation, seasonal differencing, and regular differencing has successfully removed the non-stationary features of the original series.

3.5 ACF and PACF After Differencing

The ACF and PACF of the differenced series `cardio_diff2` are shown in Figure 7. These plots are used in Section 4 to identify candidate orders for the autoregressive (AR) and moving average (MA) components of the SARIMA model.

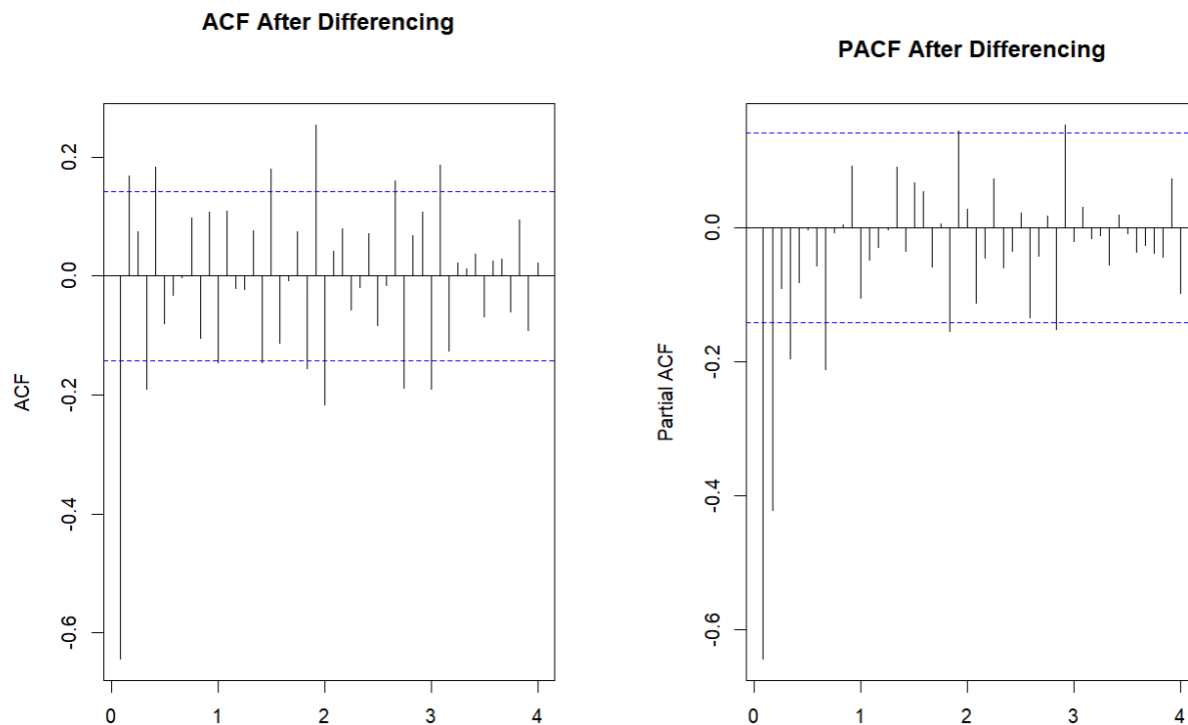


Figure 7. ACF (left) and PACF (right) of the differenced series (`cardio_diff2`).

Unlike the ACF of the original series (Figure 4), the ACF in Figure 7 no longer shows slow decay, and most autocorrelations now lie within or close to the 95% significance bounds, with a small number of isolated significant spikes (notably around lags 1, 12 and 24). This is consistent with a stationary series with some remaining short-term and seasonal dependence that can be captured by low-order AR and/or MA terms, examined formally via the EACF and BIC table in Section 4.

3.6 Stationarity Tests After Differencing

The ADF and KPSS tests were repeated on the differenced series `cardio_diff2` to confirm that stationarity had been achieved. The results are summarised in Table 3.

Test	p-value	Conclusion
ADF (differenced series)	0.01	Stationary (reject H0 of non-stationarity)
KPSS (differenced series)	0.10	Stationary (fail to reject H0 of stationarity)

Table 3. ADF and KPSS test results for the differenced series cardio_diff2.

Both tests now agree that the series is stationary: the ADF test rejects the null hypothesis of non-stationarity ($p = 0.01$), and the KPSS test fails to reject the null hypothesis of stationarity ($p = 0.10$). This confirms that the combination of a log transformation, one seasonal difference ($D = 1$, lag 12), and one regular difference applied during preprocessing is sufficient to produce a stationary series suitable for SARIMA modelling. As discussed further in Section 5, the final selected model was fitted directly to the log-transformed series cardio_ts_log with the seasonal difference ($D = 1$) specified internally by the Arima() function, and the model search determined that no additional regular differencing term ($d = 0$) was required once the seasonal difference and AR/drift terms were included.

4. Model Specification

With the series rendered stationary (Section 3), the next step is to identify suitable candidate orders for the non-seasonal AR and MA components of the SARIMA model. Consistent with the methods covered in class, model specification was carried out using the Extended Autocorrelation Function (EACF) table and the Bayesian Information Criterion (BIC) table produced by the armasubsets() function from the TSA package, both applied to the differenced series cardio_diff2.

4.1 Extended Autocorrelation Function (EACF)

The EACF table provides a systematic way of identifying plausible (p , q) combinations for an ARMA model by examining a table of generalised autocorrelations for increasing orders of autoregressive (AR, rows) and moving average (MA, columns) terms. An 'o' indicates a non-significant value (consistent with that order being adequate), while an 'x' indicates a significant value. The appropriate order is identified by locating the vertex of the largest triangle of 'o' entries. The EACF table for cardio_diff2 is shown in Figure 8.

```

> eacf(cardio_diff2)
AR/MA
  0  1  2  3  4  5  6  7  8  9 10 11 12 13
0 x x o x x o o o o o o o o o o
1 x x o x x o x o o o o o o o o
2 x x o o o o o o o o o o o o
3 x x x o o o o o o o o o o o
4 x o x o o o o o o o o o o o
5 x o o o o x o o o o o o o o
6 x o o x o o o o o o o o o o
7 x o x x o o o o o o o o o o
> |

```

Figure 8. EACF table for the differenced series (*cardio_diff2*).

Reading across the rows (AR order, $p = 0$ to 7) and down the columns (MA order, $q = 0$ to 13), a triangle of 'o' entries begins to form from around AR order 2–3 onward, with rows 2, 3, 4 and 7 showing largely 'o' entries from MA order 0 or 2 onward. This suggests that low-order AR terms (around $p = 2$ or 3) combined with a low-order MA term ($q = 0$ or 2) are likely to provide an adequate representation of the non-seasonal structure of the series. These observations were used to propose the non-seasonal orders for the candidate models in Section 4.3.

4.2 BIC Table (*armasubsets*)

As a second, complementary approach, the *armasubsets*() function was used to generate a BIC-based subset selection table, fitting models with up to 7 AR (p -lag) and 7 MA (error-lag) terms and ranking subsets of these terms by BIC. The resulting BIC table is shown in Figure 9, where darker shaded cells indicate terms that are included in the model corresponding to that row, and rows are ordered from the lowest BIC (top) to the highest BIC (bottom) among the subsets considered.

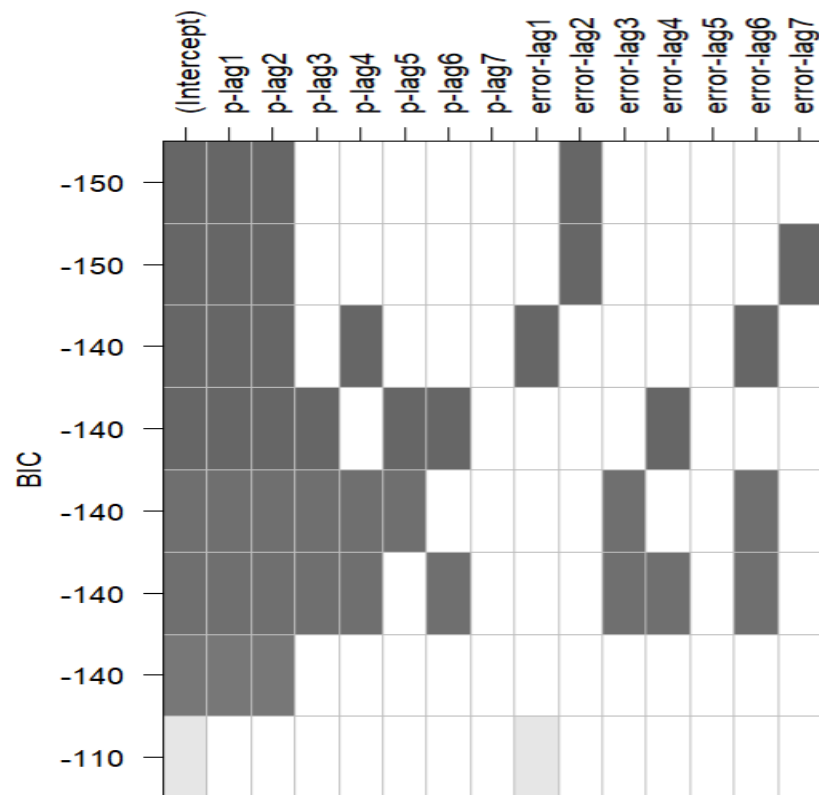


Figure 9. BIC table from `armasubsets()` for the differenced series (`cardio_diff2`), showing the AR (p -lag) and MA (error-lag) terms included in the best-performing model subsets.

The rows with the lowest BIC values consistently include the p -lag1, p -lag2 and p -lag3 terms (and, in some rows, error-lag1 and error-lag2), reinforcing the EACF-based suggestion that an AR order of around 2–3 is appropriate. Several of the better-performing subsets also include one or two seasonal moving-average-type error terms, consistent with the inclusion of a seasonal moving average (SMA) component in the final model.

4.3 Candidate Models

Based on the EACF table (Section 4.1) and the BIC table (Section 4.2), together with the seasonal differencing order $D = 1$ (lag 12) identified in Section 3, three candidate SARIMA models were specified for fitting and comparison in Section 5. All three candidate models were fitted to the log-transformed series `cardio_ts_log` (rather than to the manually differenced series `cardio_diff2`), with the seasonal difference ($D = 1$) and a drift term

specified directly within the `Arima()` function, since the differenced series in Figure 6 still showed a small but consistent non-zero mean, indicating that a drift term was required.

Candidate	Model specification	Basis for selection
Candidate 1	ARIMA(3,0,0)(0,1,2) ₁₂ with drift	EACF vertex around AR order 3; BIC table strongly favours p-lag1–p-lag3; D = 1 from Section 3.2
Candidate 2	ARIMA(1,0,0)(0,1,1) ₁₂ with drift	Simplest model consistent with EACF/BIC tables; included as a parsimonious benchmark
Candidate 3	ARIMA(2,0,0)(0,1,1) ₁₂ with drift	EACF vertex around AR order 2; intermediate complexity between Candidates 1 and 2

Table 4. Candidate SARIMA models identified from the EACF table and the BIC table.

Each of these candidate models was fitted to `cardio_ts_log` and compared using information criteria (AIC, AICc, BIC) and error measures (RMSE, MAE, MAPE, MASE) in Section 5. As an additional verification step, the `auto.arima()` function (with `stepwise = FALSE` and `approximation = FALSE`, so that all possible models within the specified order limits are evaluated) was also run on `cardio_ts_log`. This was used purely to confirm that the model identified manually using the EACF and BIC table was indeed optimal in terms of AIC, rather than as the primary model specification method.

5. Model Fitting, Selection and Parameter Estimation

5.1 Fitting the Candidate Models

Each of the three candidate models identified in Section 4.3 was fitted to the log-transformed series `cardio_ts_log` using the `Arima()` function, with the seasonal order and a drift term specified explicitly. The three fitted models were:

- **Model 1:** ARIMA(3,0,0)(0,1,2)₁₂ with drift
- **Model 2:** ARIMA(1,0,0)(0,1,1)₁₂ with drift
- **Model 3:** ARIMA(2,0,0)(0,1,1)₁₂ with drift

5.2 Model Comparison

The three candidate models were compared using the Akaike Information Criterion (AIC), corrected AIC (AICc), Bayesian Information Criterion (BIC), and a range of in-sample

accuracy measures obtained via the `accuracy()` function. The results are summarised in Table 5.

Model	AIC	AICc	BIC	RMSE	MAE	MAPE (%)	MASE
Model 1: ARIMA(3,0,0)(0,1,2) ₁₂ + drift	-523.16	-522.55	-500.36	0.0578	0.0440	0.3025	0.5828
Model 2: ARIMA(1,0,0)(0,1,1) ₁₂ + drift	-491.92	-491.70	-478.89	0.0639	0.0502	0.3463	0.6657
Model 3: ARIMA(2,0,0)(0,1,1) ₁₂ + drift	-500.09	-499.77	-483.80	0.0622	0.0484	0.3334	0.6411

Table 5. Comparison of information criteria and in-sample accuracy measures for the three candidate SARIMA models (all measures computed on the log scale).

Model 1, ARIMA(3,0,0)(0,1,2)₁₂ with drift, achieves the lowest (best) value for every criterion considered: it has the lowest AIC, AICc, and BIC, and the lowest RMSE, MAE, MAPE, and MASE among the three candidates. Model 1 was therefore selected as the final model (`best_model`) for diagnostic checking and forecasting.

5.3 Verification with `auto.arima()`

As described in Section 4.3, the `auto.arima()` function was run on `cardio_ts_log` with `stepwise = FALSE` and `approximation = FALSE` to exhaustively search across the space of candidate SARIMA models as an independent verification step. The model selected by `auto.arima()` was:

ARIMA(3,0,0)(0,1,2)₁₂ with drift, AIC = -523.16

This is identical, in both model orders and estimated coefficients, to Model 1 selected manually using the EACF table and the BIC table in Sections 4 and 5.2. This agreement between the class-taught EACF/BIC subset-selection approach and the automated search confirms that Model 1 is an appropriate and well-justified choice of model for this series.

5.4 Parameter Estimates for the Selected Model

Table 6 presents the estimated coefficients, standard errors, and approximate significance (based on whether the coefficient is more than approximately two standard errors from zero, i.e. $|\text{coefficient}| / \text{SE} > 1.96$) for the selected model, ARIMA(3,0,0)(0,1,2)₁₂ with drift, fitted to cardio_ts_log.

Parameter	Estimate	Std. Error	Significant at 5%?	Interpretation
ar1	-0.1175	0.0743	No	Weak negative association with the previous month's value
ar2	0.2540	0.0679	Yes	Positive association at a 2-month lag
ar3	0.3150	0.0732	Yes	Strongest AR term; reflects a roughly 3-month cycle
sma1	-0.1820	0.0826	Yes	Seasonal moving-average correction at lag 12
sma2	-0.1791	0.0703	Yes	Additional seasonal moving-average correction at lag 24
drift	0.0051	0.0004	Yes	Captures the consistent long-run upward trend on the log scale

Table 6. Parameter estimates for the selected model, ARIMA(3,0,0)(0,1,2)₁₂ with drift (Series: cardio_ts_log).

The model has a residual variance of $\sigma^2 = 0.003662$ (log likelihood = 268.58). While the ar1 coefficient is not individually statistically significant at the 5% level, it was retained because (i) it is part of the AR(3) structure jointly identified as optimal by both the EACF/BIC subset-selection approach and the independent auto.arima() search (Section 5.3), and (ii) removing it would change the model order away from the jointly-selected specification without improving the overall information criteria reported in Table 5. The drift coefficient of 0.0051 (on the log scale) corresponds to an average monthly growth rate of approximately 0.51% in cardiovascular prescription volumes, consistent with the long-term upward trend identified in Section 2.

6. Diagnostic Checking

Before using the selected model (Model 1: ARIMA(3,0,0)(0,1,2)₁₂ with drift) for forecasting, its residuals were examined to check whether the model adequately captures

the structure of the series. If the model is well-specified, the residuals should behave approximately like white noise: uncorrelated, with approximately zero mean and constant variance, and approximately normally distributed.

6.1 Residual Plots

Figure 10 presents three diagnostic plots produced by the `checkresiduals()` function: a time plot of the residuals, the ACF of the residuals, and a histogram of the residuals with a fitted normal density curve overlaid.

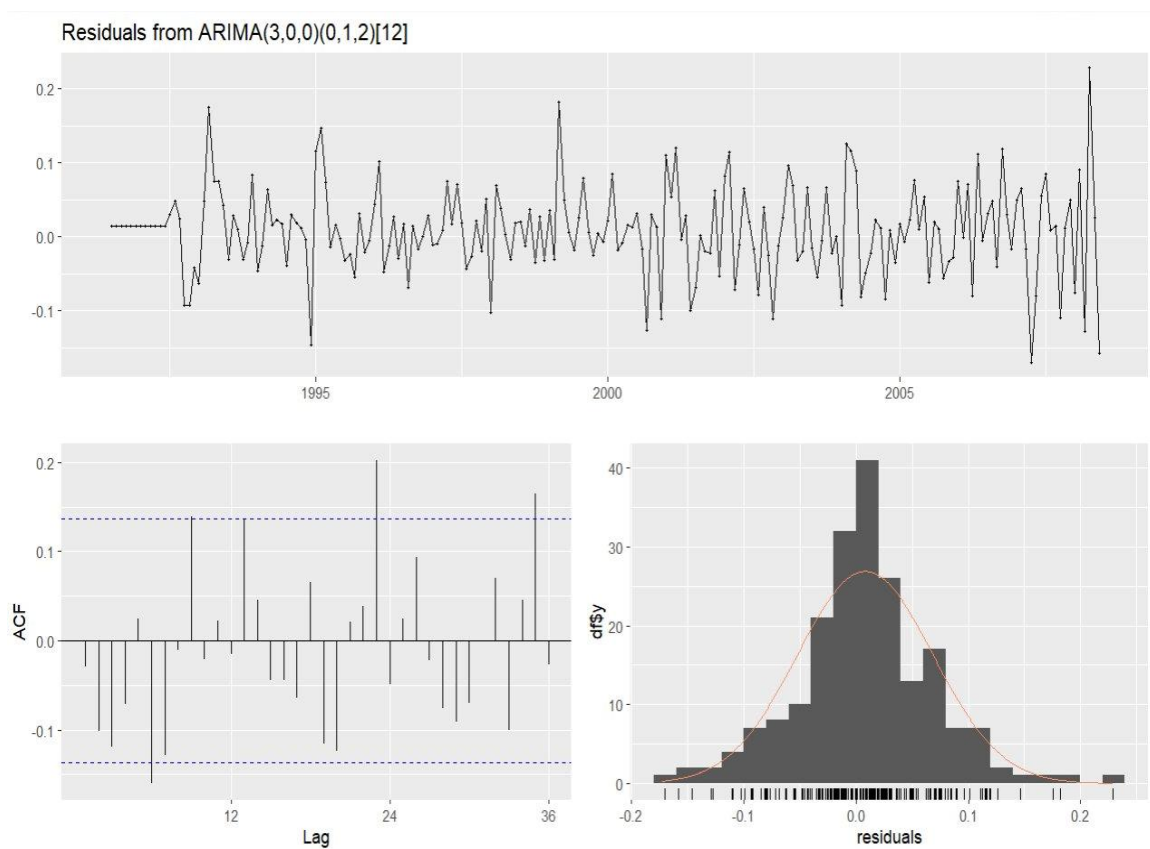


Figure 10. Residual diagnostics for the selected model: time plot of residuals (top), ACF of residuals (bottom left), and histogram of residuals with fitted normal curve (bottom right).

The time plot of the residuals (top panel) shows no obvious remaining trend or seasonal pattern, and the residuals appear to fluctuate randomly around zero with reasonably constant variance, with the exception of a small number of larger spikes (for example, around 1999 and at the very end of the series). The ACF of the residuals (bottom left panel) shows that the vast majority of autocorrelations lie within the 95% significance bounds, with only isolated spikes (around lags 11, 23 and 35) exceeding the bounds. The

histogram (bottom right panel) shows a roughly bell-shaped distribution centred close to zero, broadly consistent with the overlaid normal curve, although with a slightly heavier right tail.

6.2 Ljung-Box Test for Residual Autocorrelation

The Ljung-Box test was used to formally test for remaining autocorrelation in the residuals, with the null hypothesis (H_0) that the residuals are independently distributed (i.e. the model has captured all the autocorrelation structure in the series). Two versions of this test were performed, and both are reported here because they use different degrees-of-freedom adjustments, which is important to understand when interpreting the results.

First, the `checkresiduals()` function automatically performs a Ljung-Box test with the degrees of freedom adjusted for the number of AR and MA parameters estimated in the model (`fitdf = 5`, corresponding to `ar1`, `ar2`, `ar3`, `sma1` and `sma2`):

$Q^* = 32.898$, $df = 19$, $p\text{-value} = 0.0247$

Second, a manual `Box.test()` was performed with `lag = 24` and `type = "Ljung-Box"` without a degrees-of-freedom adjustment for estimated parameters (`df = 24`):

$Q^* = 32.898$, $df = 24$, $p\text{-value} = 0.1062$

Both versions of the test produce the same test statistic ($Q^* = 32.898$), since this depends only on the residual autocorrelations and not on the degrees-of-freedom adjustment; the difference lies entirely in the p-value, which depends on the degrees of freedom used for the reference chi-squared distribution. Using the more conservative, parameter-adjusted version (`df = 19`), the p-value of 0.0247 is below the conventional 0.05 threshold, suggesting that some statistically detectable autocorrelation remains in the residuals. Using the unadjusted version (`df = 24`), the p-value of 0.1062 is above 0.05, suggesting no significant remaining autocorrelation.

Taken together with the residual ACF plot in Figure 10 – which shows only a small number of isolated significant spikes, all at relatively long lags (11, 23, 35) rather than at short lags – this borderline result indicates that any remaining autocorrelation in the residuals

is minor and is likely attributable to a small number of irregular observations (for example, the larger residual spikes visible around 1999 and towards the end of the series, which may correspond to PBS policy or scheme changes affecting prescription volumes in those years) rather than to a fundamental misspecification of the model order. Combined with the very strong in-sample accuracy measures reported in Table 5 (MAPE = 0.30%, MASE = 0.58), Model 1 is considered adequate for the purpose of generating short-term forecasts, while this minor residual autocorrelation is noted as a limitation of the model in Section 8.2.

6.3 Shapiro-Wilk Normality Test

The Shapiro-Wilk test was used to assess the normality of the model residuals, with the null hypothesis (H_0) that the residuals are drawn from a normal distribution. The test produced the following result:

$W = 0.9844$, $p\text{-value} = 0.024$

With $p = 0.024$, the null hypothesis of normality is formally rejected at the 5% significance level. However, the test statistic $W = 0.9844$ is very close to 1 (its maximum possible value for a perfectly normal sample), indicating that the departure from normality is minor in practical terms. The Shapiro-Wilk test is also known to be highly sensitive to small departures from normality when the sample size is reasonably large ($n = 204$ in this case), meaning that even slight asymmetries or heavier tails – as suggested by the histogram in Figure 10 – can produce a statistically significant result. Given the very high value of W , the minor departure from normality is not considered to materially affect the validity of the point forecasts or the general shape of the prediction intervals presented in Section 7, although the prediction intervals should be interpreted as approximate.

6.4 Overall Diagnostic Conclusion

Overall, the diagnostic checks indicate that the selected model (Model 1: ARIMA(3,0,0)(0,1,2)₁₂ with drift) provides a reasonable fit to the data. The residuals show no obvious remaining trend or seasonal structure, the majority of residual autocorrelations are within acceptable bounds, and although both the Ljung-Box test (under one degrees-of-freedom specification) and the Shapiro-Wilk test return p-values below 0.05, the

magnitudes of the test statistics ($Q^* = 32.898$ against a benchmark of approximately 30–36 depending on degrees of freedom, and $W = 0.9844$) indicate that these departures from the ideal white-noise/normality assumptions are minor. Combined with the strong in-sample accuracy of the model (Table 5), Model 1 is considered fit for the purpose of producing the 10-month forecast presented in Section 7.

7. Forecasting

Using the selected model (Model 1: $ARIMA(3,0,0)(0,1,2)_{12}$ with drift, fitted to `cardio_ts_log`), point forecasts and 80% and 95% prediction intervals were generated for the 10 months following the end of the observed series, from July 2008 to April 2009, using the `forecast()` function with $h = 10$. Because the model was fitted to the log-transformed series, all forecasts and prediction interval bounds were back-transformed to the original scale (number of scripts) by exponentiation.

7.1 10-Month Forecast Table

Table 7 presents the point forecasts together with the 95% prediction intervals for each of the 10 forecast months, on the original (back-transformed) scale.

Month	Point Forecast	95% PI Lower	95% PI Upper
Jul 2008	4,071,418	3,616,085	4,584,085
Aug 2008	3,322,095	2,948,157	3,743,461
Sep 2008	2,909,601	2,571,399	3,292,285
Oct 2008	2,675,778	2,356,231	3,038,662
Nov 2008	2,412,998	2,124,832	2,740,243
Dec 2008	2,159,429	1,899,235	2,455,271
Jan 2009	2,259,718	1,987,011	2,569,853
Feb 2009	4,107,516	3,611,627	4,671,493
Mar 2009	3,864,591	3,397,380	4,396,053
Apr 2009	4,560,805	4,009,332	5,188,132

Table 7. Point forecasts and 95% prediction intervals for cardiovascular prescription scripts, July 2008 – April 2009 (back-transformed to the original scale).

7.2 Forecast Plot

Figure 11 displays the historical series alongside the 10-month forecast, with shaded regions representing the 80% and 95% prediction intervals.

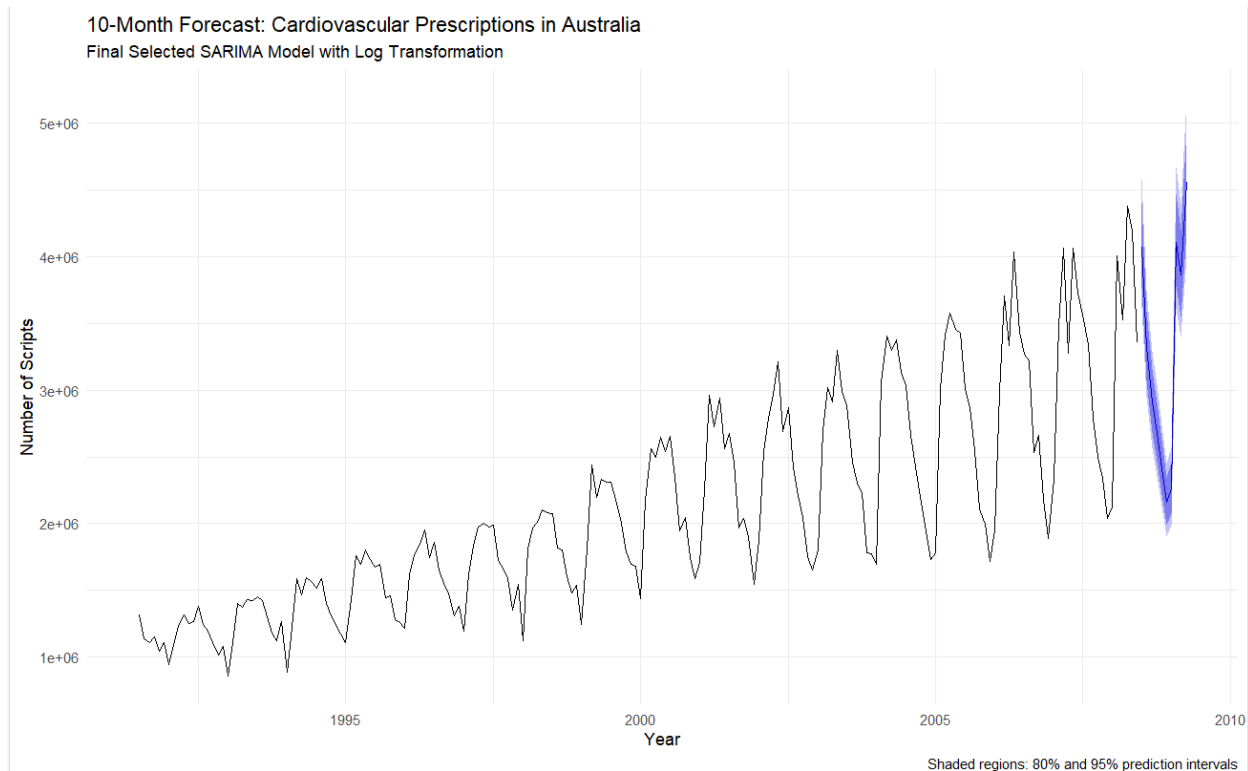


Figure 11. 10-month forecast of monthly cardiovascular prescription scripts (July 2008 – April 2009), with 80% and 95% prediction intervals (shaded).

7.3 Discussion of Forecast Results

The forecast results in Table 7 and Figure 11 show that the seasonal pattern identified throughout this report is expected to continue into the forecast period. Prescription volumes are forecast to decline from a relatively high level in July 2008 (4,071,418 scripts) to a low point around December 2008 (2,159,429 scripts), before rising sharply again to a peak in April 2009 (4,560,805 scripts) – the highest value across the entire forecast horizon. This pattern of an autumn/winter trough followed by a rise to a peak in the following autumn is consistent with the seasonal cycle identified in Sections 2.1 and 2.4, where prescription volumes typically peak between February and April and reach a trough between July and September.

The forecast also reflects the long-term upward trend identified throughout the report: the forecast peak of 4,560,805 scripts in April 2009 is higher than any value observed in the historical data (maximum of 4,377,361 scripts, Table 1), consistent with the positive drift term (0.0051 on the log scale, Table 6) estimated for the model. The 95% prediction intervals widen progressively as the forecast horizon increases, reflecting the increasing uncertainty associated with forecasting further into the future; nonetheless, the intervals remain reasonably tight relative to the scale of the series (for example, the April 2009 interval of approximately 4.01 million to 5.19 million scripts represents a relative width of around 26% of the point forecast), indicating that the model produces informative forecasts over this 10-month horizon.

8. Conclusion

8.1 Summary of Findings

This report applied a SARIMA modelling framework to monthly PBS cardiovascular prescription data (Co-payments, Concessional) for Australia, covering July 1991 to June 2008 (204 observations). Exploratory analysis revealed a strong upward trend, a consistent annual seasonal cycle (peaking February–April and troughing July–September), and increasing variance proportional to the level of the series, motivating a multiplicative decomposition and a log transformation.

Formal ADF and KPSS tests, together with the `ndiffs()` and `nsdiffs()` functions, indicated that one seasonal difference (lag 12) and one regular difference were required to achieve stationarity; after applying a log transformation and this differencing during preprocessing, both tests confirmed that the resulting series was stationary. Model orders for the non-seasonal AR and MA components were identified using the EACF table and the BIC table from `armasubsets()`, in line with the methods covered in class, leading to three candidate models. Of these, $\text{ARIMA}(3,0,0)(0,1,2)_{12}$ with drift (Model 1) achieved the lowest AIC, AICc, BIC, RMSE, MAE, MAPE and MASE, and was independently confirmed as optimal by an exhaustive `auto.arima()` search.

Diagnostic checking showed that the residuals of Model 1 were broadly consistent with white noise, with only minor and borderline evidence of remaining autocorrelation and a

very slight departure from normality, neither of which materially compromises the model's strong in-sample accuracy (MAPE = 0.30%, MASE = 0.58). Using this model, a 10-month forecast (July 2008 – April 2009) was produced, predicting that cardiovascular prescription volumes will follow the established seasonal pattern – declining to a trough around December 2008 (approximately 2.16 million scripts) before rising to a new peak in April 2009 (approximately 4.56 million scripts), the highest value in the series to date, consistent with the model's positive estimated drift and the long-term upward trend in cardiovascular prescriptions in Australia.

8.2 Limitations and Future Work

- **Residual autocorrelation:** The Ljung-Box test produced a borderline result ($p = 0.0247$ with $df = 19$, adjusted for estimated parameters), indicating some minor remaining autocorrelation in the residuals, likely associated with a small number of irregular years in the data. Future work could explore models with additional seasonal AR terms or intervention/outlier modelling to address these specific periods.
- **Univariate model:** The model relies solely on the historical pattern of the series itself. External factors such as changes in PBS policy, co-payment thresholds, population ageing rates, or the introduction of new cardiovascular medicines are not explicitly modelled, but could be incorporated as exogenous regressors in a dynamic regression or ARIMAX framework.
- **Forecast horizon:** The 10-month forecast horizon was specified by the assignment brief. Prediction intervals widen with longer horizons, so forecasts beyond this period should be treated with increasing caution and the model should be periodically re-estimated as new data becomes available.
- **Normality of residuals:** The Shapiro-Wilk test indicated a statistically significant, though practically minor, departure from normality ($W = 0.9844$). This means the reported prediction intervals, which assume normally distributed errors, are approximate; bootstrap-based prediction intervals could be explored as a more robust alternative in future work.

8.3 Practical Implications

The forecasts produced in this report could assist health policy makers and PBS administrators in anticipating demand for cardiovascular medicines over the short term,

supporting decisions around pharmaceutical procurement, budget allocation, and the planning of subsidy schemes. The clear seasonal pattern – with demand peaking in the autumn months and dipping in winter/spring – may also be useful for informing the timing of prescription renewal reminders or stock management strategies for pharmacies and suppliers.

9. References

1. Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723. <https://doi.org/10.1109/TAC.1974.1100705>
2. Australian Government Department of Health and Aged Care. (2024). *Pharmaceutical Benefits Scheme (PBS)*. <https://www.pbs.gov.au/pbs/home>
3. Australian Institute of Health and Welfare. (2023). *Heart, stroke and vascular disease: Australian facts*. <https://www.aihw.gov.au/reports/heart-stroke-vascular-diseases/hsvd-facts>
4. Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time series analysis: Forecasting and control* (5th ed.). John Wiley & Sons.
5. Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366), 427–431. <https://doi.org/10.2307/2286348>
6. Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>
7. Hyndman, R. J., & Khandakar, Y. (2008). Automatic time series forecasting: The forecast package for R. *Journal of Statistical Software*, 27(3), 1–22. <https://doi.org/10.18637/jss.v027.i03>
8. Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., & Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1–3), 159–178. [https://doi.org/10.1016/0304-4076\(92\)90104-Y](https://doi.org/10.1016/0304-4076(92)90104-Y)
9. Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303. <https://doi.org/10.1093/biomet/65.2.297>
10. O'Hara-Wild, M., Hyndman, R., Wang, E., Cannoodt, R., Chhay, L., Cooper, J., & Hayes, S. (2024). *tsibbledata: Diverse datasets for 'tsibble' (R package)*. <https://CRAN.R-project.org/package=tsibbledata>

11. R Core Team. (2024). *R: A language and environment for statistical computing* (Version 4.4.3) [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>
12. Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2), 461–464. <https://doi.org/10.1214/aos/1176344136>
13. Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3/4), 591–611. <https://doi.org/10.2307/2333709>

10. Appendix A: Complete R Code

The complete R code used to perform the data loading, exploratory analysis, stationarity testing, transformation, differencing, model specification (EACF and BIC table), model fitting and comparison, diagnostic checking, and forecasting reported in this document is provided below.

```
library(tsibbledata)
library(dplyr)
library(forecast)
library(tseries)
library(ggplot2)
library(TSA)

# Data Loading
my_data <- PBS %>%
  filter(ATC1 == "C",           # Cardiovascular system
         Type == "Co-payments",
         Concession == "Concessional") %>%
  group_by(Month) %>%
  summarise(Scripts = sum(Scripts)) %>%
  ungroup()

# To View it
View(my_data)

# Convert to ts object
cardio_ts <- ts(my_data$Scripts,
               start = c(1991, 7),
               frequency = 12)

# Time plot
autoplot(cardio_ts) +
  labs(title = "Monthly Cardiovascular Prescriptions in Australia",
       subtitle = "PBS Medicare Data: July 1991 - June 2008",
       x = "Year",
       y = "Number of Scripts") +
  theme_minimal()

# DESCRIPTIVE STATISTICS
summary(cardio_ts)
cat("Mean:", mean(cardio_ts), "\n")
cat("Std Dev:", sd(cardio_ts), "\n")
```

```

cat("Min:", min(cardio_ts), "\n")
cat("Max:", max(cardio_ts), "\n")

# DECOMPOSITION
# Multiplicative decomposition (because variance increases over time)
cardio_decomp <- decompose(cardio_ts, type = "multiplicative")
plot(cardio_decomp)

# Seasonal plot
ggseasonplot(cardio_ts,
              year.labels = TRUE,
              main = "Seasonal Plot: Cardiovascular Scripts") +
  theme_minimal()

# ACF AND PACF (ORIGINAL SERIES)
par(mfrow = c(1, 2))
acf(cardio_ts, lag.max = 48,
    main = "ACF - Cardiovascular Scripts")
pacf(cardio_ts, lag.max = 48,
    main = "PACF - Cardiovascular Scripts")

# STATIONARITY TESTS (ORIGINAL SERIES)
adf.test(cardio_ts)
kpss.test(cardio_ts)

# HOW MANY DIFFERENCES NEEDED
ndiffs(cardio_ts)
nsdiffs(cardio_ts)

# TRANSFORMATION & DIFFERENCING
# Log transformation
cardio_ts_log <- log(cardio_ts)

autoplot(cardio_ts_log) +
  labs(title = "Log Transformed Cardiovascular Scripts",
       x = "Year", y = "Log Scripts") +
  theme_minimal()

# Seasonal differencing
cardio_diff <- diff(cardio_ts_log, lag = 12)

# Regular differencing
cardio_diff2 <- diff(cardio_diff, differences = 1)

# Plot differenced series
autoplot(cardio_diff2) +
  labs(title = "Differenced Series (Seasonal + Regular)",
       x = "Year", y = "Differenced Scripts") +
  theme_minimal()

# ACF/PACF after differencing
par(mfrow = c(1, 2))
acf(cardio_diff2, lag.max = 48,
    main = "ACF After Differencing")
pacf(cardio_diff2, lag.max = 48,
    main = "PACF After Differencing")

# Stationarity after differencing
adf.test(cardio_diff2)
kpss.test(cardio_diff2)

# EACF TABLE
eacf(cardio_diff2)

```

```

# BIC TABLE
res = armasubsets(y = cardio_diff2, nar = 7, nma = 7,
                  y.name = 'p', ar.method = 'ml')
plot(res)

# CANDIDATE MODELS (based on EACF and BIC table)
modell1 <- Arima(cardio_ts_log, order = c(3,0,0),
                seasonal = list(order = c(0,1,2), period = 12),
                include.drift = TRUE)
summary(modell1)
checkresiduals(modell1)

modell2 <- Arima(cardio_ts_log, order = c(1,0,0),
                seasonal = list(order = c(0,1,1), period = 12),
                include.drift = TRUE)
summary(modell2)

modell3 <- Arima(cardio_ts_log, order = c(2,0,0),
                seasonal = list(order = c(0,1,1), period = 12),
                include.drift = TRUE)
summary(modell3)

# MODEL COMPARISON
AIC(modell1, modell2, modell3)
BIC(modell1, modell2, modell3)

accuracy(modell1)
accuracy(modell2)
accuracy(modell3)

# AUTO.ARIMA - VERIFICATION ONLY
auto_model <- auto.arima(cardio_ts_log,
                        stepwise = FALSE,
                        approximation = FALSE,
                        trace = TRUE)

summary(auto_model)

# SELECTING FINAL MODEL
best_model <- modell1

# RESIDUAL DIAGNOSTICS
checkresiduals(best_model)

# Ljung-Box test
Box.test(best_model$residuals, lag = 24, type = "Ljung-Box")

# Normality test of residuals
shapiro.test(best_model$residuals)

# Residual ACF
par(mfrow = c(1,1))
acf(best_model$residuals, main = "ACF of Residuals")

# ACCURACY
accuracy(best_model)

# FORECASTING
forecast_log <- forecast(best_model, h = 10, level = c(80, 95))
print(forecast_log)

# Back-transform to original scale
forecast_final <- forecast_log
forecast_final$mean <- exp(forecast_log$mean)
forecast_final$lower <- exp(forecast_log$lower)

```

```

forecast_final$upper <- exp(forecast_log$upper)
forecast_final$x      <- exp(forecast_log$x)

# Forecast table
forecast_table <- data.frame(
  Month = c("Jul 2008", "Aug 2008", "Sep 2008",
            "Oct 2008", "Nov 2008", "Dec 2008",
            "Jan 2009", "Feb 2009", "Mar 2009", "Apr 2009"),
  Point_Forecast = round(as.numeric(forecast_final$mean)),
  Lo_80 = round(as.numeric(forecast_final$lower[,1])),
  Hi_80 = round(as.numeric(forecast_final$upper[,1])),
  Lo_95 = round(as.numeric(forecast_final$lower[,2])),
  Hi_95 = round(as.numeric(forecast_final$upper[,2]))
)
print(forecast_table)

# Final forecast plot
autoplot(forecast_final) +
  labs(title = "10-Month Forecast: Cardiovascular Prescriptions in Australia",
       subtitle = "Final Selected SARIMA Model with Log Transformation",
       x = "Year",
       y = "Number of Scripts",
       caption = "Shaded regions: 80% and 95% prediction intervals") +
  theme_minimal()

```